Final Project Technical Report

Bank Queuing System

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1 About the Project

Our project presents a C++ simulation of a bank queuing system. It models customer arrivals, service processes, and teller operations to evaluate the system efficiency and performance of the Bank.

2 System Idea

The program simulates a real bank environment where customers arrive at different times, request various services, and are served by available tellers. The system focuses on analyzing waiting time, service efficiency, and teller utilization.

3 Program Structure

The system is organized into four main modules:

- **Configuration** → defines system parameters.
- **Simulation** → executes the queuing process.
- **Reports** → displays detailed results.
- **Statistics** → summarizes overall performance.

This modular design improves clarity, maintainability, and ease of testing.

4 System Workflow and Simulation Logic

The system begins with user configuration, where the number of tellers, working hours, transaction types, and yearly customer target are defined. These inputs determine how the simulation behaves.

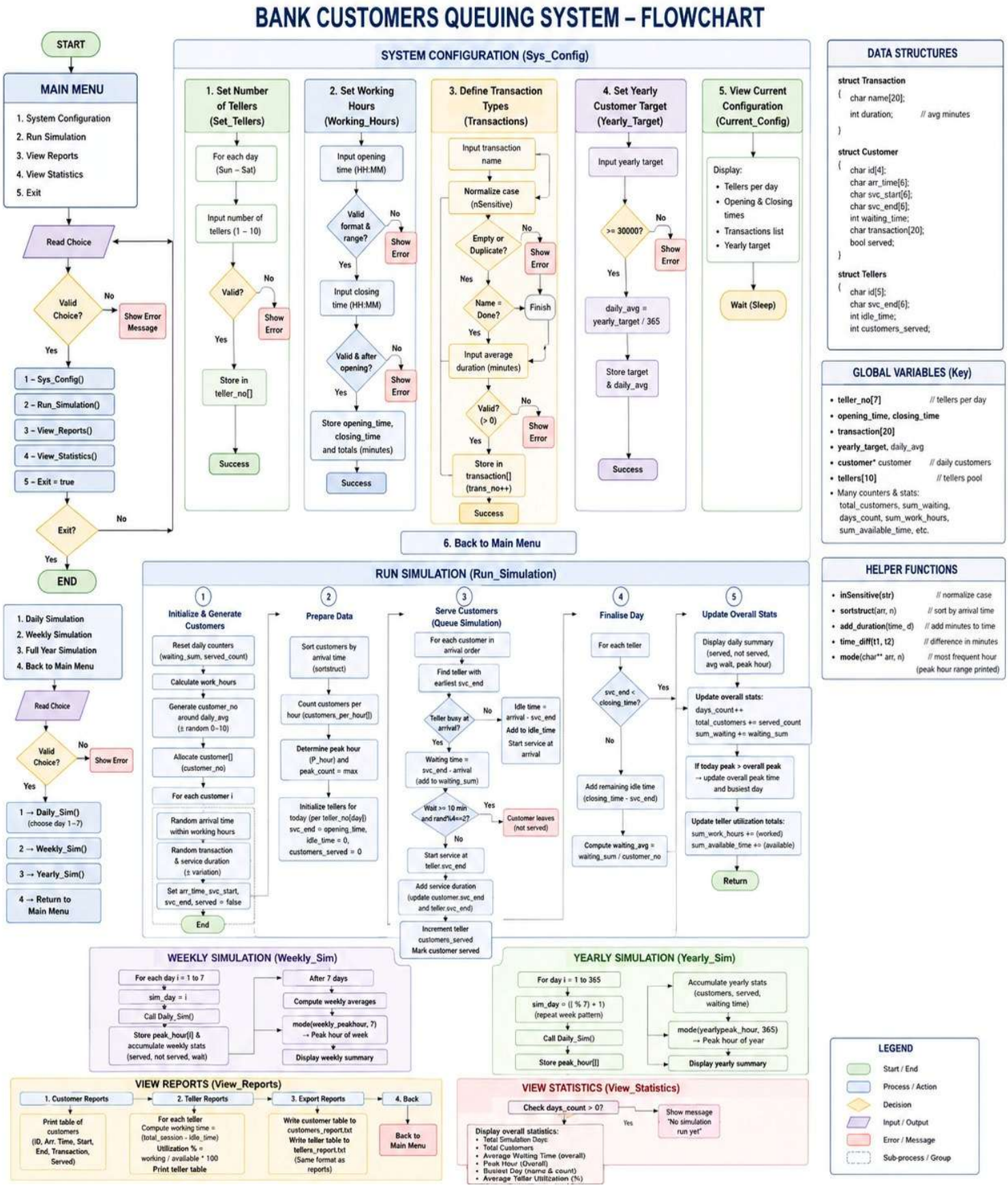
During simulation, customers are generated based on the daily average derived from the yearly target. Each customer is assigned a random arrival time within working hours and a transaction type that defines their service duration.

Customers are processed using the **First-Come First-Served (FCFS)** approach. When a teller becomes available, the next customer in the queue is served; otherwise, customers wait in line. For each customer, the system determines service start and end times and checks whether they are served before closing time.

Throughout the process, the system calculates waiting times and tracks teller activity, including working and idle periods. After all customers are processed, the system produces reports for both customers and tellers, then computes overall statistics such as average waiting time, peak hours, and utilization rates.

5 Flowchart

The system follows a simple execution flow:



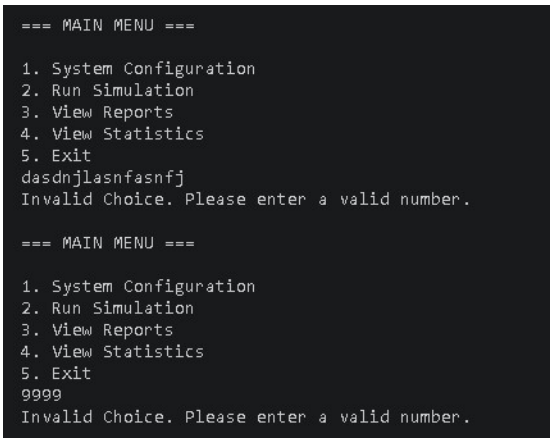
6 Validation and Error Handling

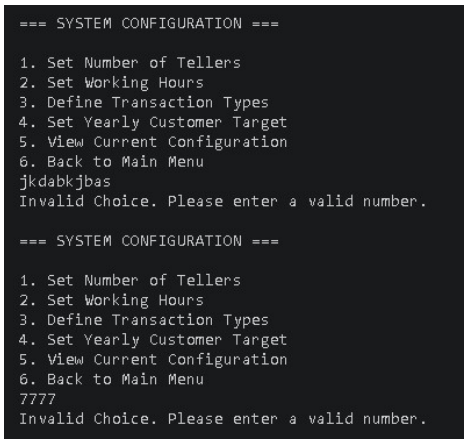
The system ensures correctness through input validation and error handling:

- Invalid menu choices are rejected.
- Inputs are checked for correct format and logical consistency.
- Constraints (e.g., time ranges, limits) are enforced.

6.1 Main menu and System Configuration

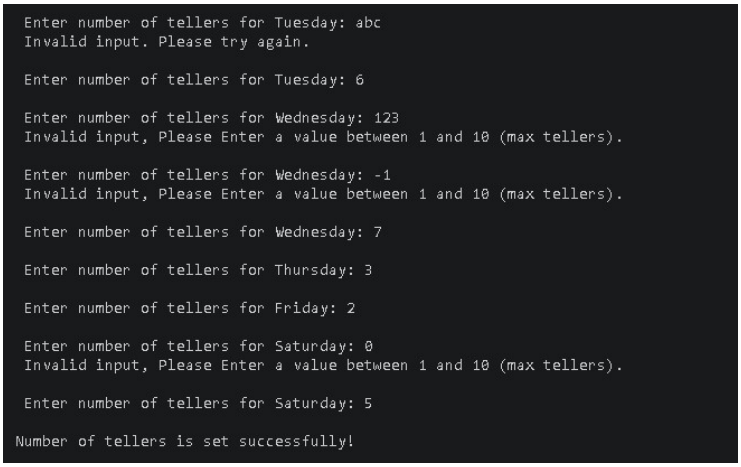
Both the main menu and the system configuration menu implement the same validation approach to ensure correct user input. The program first checks whether the entered value is a valid integer using input stream validation. If the input is invalid (such as entering characters instead of numbers), the error state is cleared, and the invalid input is discarded to prevent program failure. After confirming the input type, the system verifies that the selected option falls within the available menu range. If the value is outside the valid options, an error message is displayed, and the user is prompted to enter a correct choice. This approach ensures robust interaction and prevents unexpected behavior during execution.





6.2 Set the number of tellers.

The validation in this section ensures that the number of tellers entered is a valid positive integer and does not exceed the predefined maximum limit. The program first verifies that the input is of the correct data type to prevent runtime errors. Once the input is confirmed to be valid, it is further checked against the required constraints. If the value satisfies all conditions, it is stored; otherwise, the user is prompted to re-enter a valid number.



6.3 Set Working Hours

The working hours section validates both the format and logical correctness of the entered times. The program first ensures that the input follows the required **HH:MM** format by checking the string length and the position of the colon. It then extracts the hour and minute values and verifies that they fall within valid ranges (00–23 for hours and 00–59 for minutes). Finally, the system checks that the closing time is strictly later than the opening time. If any of these conditions fail, the user is prompted to re-enter the time.

```
Enter the opening hours in HH:MM, in 24-hour format: 05:123
Invalid opening time. Please enter time in HH:MM format.

Enter the opening hours in HH:MM, in 24-hour format: 050:23
Invalid opening time. Please enter time in HH:MM format.

Enter the opening hours in HH:MM, in 24-hour format: @3:7$
Invalid opening time! Enter a valid time, Hours (00-23) & minutes (00-59).

Enter the opening hours in HH:MM, in 24-hour format: 32:32
Invalid opening time! Enter a valid time, Hours (00-23) & minutes (00-59).

Enter the opening hours in HH:MM, in 24-hour format: 12:77
Invalid opening time! Enter a valid time, Hours (00-23) & minutes (00-59).

Enter the opening hours in HH:MM, in 24-hour format: 09:00

Opening time is set successfully!

Enter the closing hours in HH:MM, in 24-hour format: 08:30
Closing time must be after opening time.

Enter the closing hours in HH:MM, in 24-hour format: 15:30

Closing time is set successfully!
```

6.4 Define Transaction Types

In this section, the system validates transaction names and durations to ensure data consistency. The program removes leading and trailing spaces and converts input to lowercase to maintain uniformity. It checks that the transaction name is not empty and ensures uniqueness by comparing it with previously entered transactions. The keyword **"done"** is used

```
Enter transaction name (or 'done' to finish):
Transaction name should not be empty.

Enter transaction name (or 'done' to finish): deposit
Enter average duration in minutes: 3
Transaction added successfully!

Enter transaction name (or 'done' to finish): Deposit
Transaction name must be unique.

Enter transaction name (or 'done' to finish):     deposit
Transaction name must be unique.

Enter transaction name (or 'done' to finish): New Account
Enter average duration in minutes: abc
Invalid input, enter a new duration:

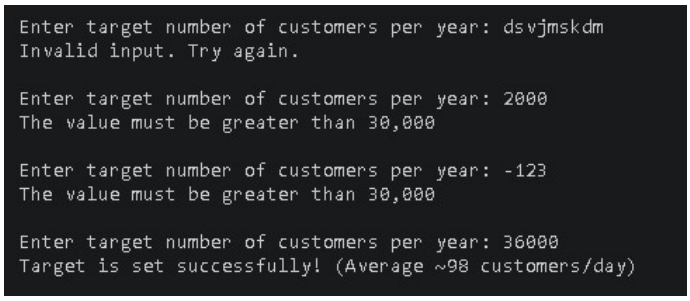
Enter average duration in minutes: -5
Duration must be positive.
Enter average duration in minutes: 7
Transaction added successfully!

Enter transaction name (or 'done' to finish): DONE
```

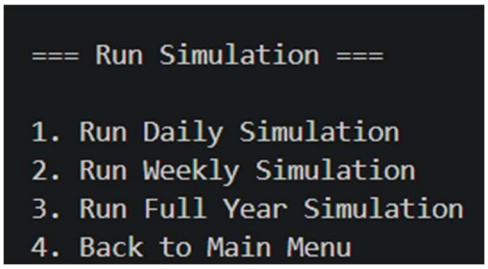
to terminate input. For the transaction duration, the program verifies that the input is a valid integer and that it is a positive value. Invalid entries result in error messages and re-prompting.

1.1 Set Yearly Customer Target

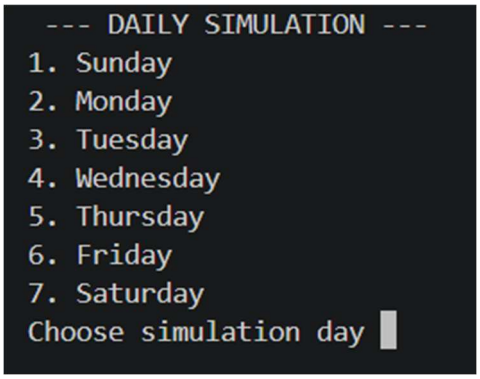
The yearly target input is validated to ensure that it is a numeric value and meets the minimum required threshold. The program checks that the input is an integer and clears any invalid input from the buffer. It then verifies that the value is greater than 30,000. If the condition is satisfied, the value is stored and used to calculate the average daily customers; otherwise, the user is prompted to enter a valid target.



After the system configuration is complete, the program transitions to the simulation stage where the user is presented with several processing scales. The user is given a chance to select any option from the run simulation menu, including (1) Run Daily Simulation, (2) Run Weekly Simulation, (3) Run Full Year Simulation, or (4) Back to Main Menu.



If the user chooses to run a daily simulation by selecting option (1), the program instantly brings the user to the day selection sub-menu. Once a specific simulation is executed and the results are displayed, the user is again presented with these options to perform further analysis or return to the primary system controls.



After selecting the desired day, the program initiates the data processing sequence for that specific 24-hour period. Once the simulation for the chosen day is complete, the system displays the resulting performance metrics and instantly brings the user back to the daily simulation menu.
The user is then again presented with the option to select a different day of the week to analyze, ranging from Sunday to Saturday. If the user has finished reviewing the daily data, they can choose to navigate back to the higher-level simulation menu to run broader weekly or yearly reports.

```
Simulating Sunday...
Customers generated: 949
Processing customers...
[==============================] 100%

 === Daily Summary ===
Customers Served: 742
Customers Not Served: 207
Average Waiting Time: 340.78
Peak Hour: 14:00 - 15:00 (174 customers)
```

Once the specific day is selected, the program executes the processing logic. The system generates a customer count and displays a progress bar while processing the simulation. After the task reaches 100%, a daily summary is provided, including metrics such as customers served, customers not served, average waiting time, and the identified peak hour.
The program then instantly brings the user back to the run simulation menu. The user is then again presented with the option to run additional simulations or return to the main menu. If the user chooses to run another daily simulation, they can repeat the process to compare data from different days of the week.

```
 === Weekly Summary ===
Customers Served: 5052
Customers Not Served: 1676
Average Waiting Time: 305.532
Peak Hour: 09:00 - 10:00 (most frequent)
```

When the user selects the weekly simulation option, the program processes the data for the entire seven-day period and displays a weekly summary as shown in image_e9d8b5.png. This report aggregates performance data, providing totals for customers served and not served, the average waiting time across the week, and the most frequent peak hour observed.

The program then instantly brings the user back to the run simulation menu. The user is then again presented with the options to run further simulations, such as a full-year analysis, or return to the main menu. If the user chooses to run the weekly simulation again, they can re-evaluate the system's efficiency over that extended timeframe, however if a full-year simulation is picked, the user is presented with a yearly summary.

```
=== YEARLY SUMMARY ===
Customers Generated: 349446
Customers Served: 263135
Customers Not Served: 86311
Average Waiting Time: 305.181 minutes
Peak Hour: 11:00 - 12:00 (most frequent)
```

After the simulation stage is completed, the user may proceed to the reporting section by selecting the “View Reports” option from the main menu. The program then displays a secondary menu containing four available options: (1) View Customer Reports, (2) View Teller Reports, (3) Export Reports to Files, and (4) Back to Main Menu. This reporting module allows the user to inspect the simulation output in a more detailed and organized manner.

```
=== VIEW REPORTS ===

1. View customer Reports
2. View Teller Reports
3. Export Reports To Files
4. Back to Main Menu
```

If the user selects the customer reports option, the system generates and displays a detailed table containing the processed customer information from the most recent simulation. The report includes each customer's identification number, arrival time, service start time, service end time, transaction type, and service status indicating whether the customer was successfully served or left the queue before receiving service. The data is displayed sequentially according to the First-Come, First-Served (FCFS) queuing policy implemented in the simulation logic. Once the report is displayed, the user may return to the reports menu to access other reporting options.

```
ID Arv Time Svc Start Svc End Transaction Served
001     09:04   09:04   09:10   Deposit Yes
-----------------------------
002     09:06   09:06   09:13   Withdraw        Yes
-----------------------------
003     09:06   09:06   09:12   Deposit Yes
-----------------------------
004     09:12   09:12   09:20   Withdraw        Yes
-----------------------------
005     09:16   09:16   09:21   Deposit Yes
-----------------------------
006     09:23   09:23   09:29   Deposit Yes
-----------------------------
007     09:28   09:28   09:35   Withdraw        Yes
-----------------------------
008     09:29   09:29   09:36   Withdraw        Yes
-----------------------------
009     09:38   09:38   09:48   Loan payment    Yes
```

When the teller reports option is selected, the system generates a detailed operational report for all tellers participating in the simulation. The report includes teller identification numbers, the number of customers served by each teller, idle time duration, total working time, and utilization percentage. The utilization percentage is calculated based on the ratio between the teller's active working time and the total available working period. This report enables the user to evaluate teller efficiency and identify periods of underutilization or excessive workload within the system.

```
Teller ID Customers Idle Working time Utilization
T001    37      68 min  292min  81.1111%
-----------------------------
T002    33      81 min  279min  77.5%
-----------------------------
T003    38      76 min  284min  78.8889%
-----------------------------
T004    12      244 min 116min  32.2222%
-----------------------------
```

If the user chooses to export reports to files, the program automatically creates external text files containing the generated reports. Two separate files are produced: “customers_report.txt” and “tellers_report.txt”. The customer report file contains all customer-related simulation data, while the teller report file contains teller activity and performance statistics. The reports are formatted into aligned columns using output formatting functions to improve readability. Once the export process is completed successfully, the program displays a confirmation message informing the user that the files have been generated and saved.

```
Customers report exported to customers_report.txt
Tellers report exported to tellers_report.txt
```

After completing any report operation, the user is automatically returned to the reports menu, where they may choose to view additional reports, export data again, or navigate back to the main menu. This looping structure allows continuous access to reporting features without restarting the program.

```
=== MAIN MENU ===

1. System Configuration
2. Run Simulation
3. View Reports
4. View Statistics
5. Exit
```

In addition to the reports module, the system provides a statistics section accessible from the main menu through the “View Statistics” option. This section presents cumulative performance statistics collected from all completed simulation sessions. Before displaying the statistics, the system first verifies whether at least one simulation has been executed.

If no simulation data exists, the program displays a warning message indicating that statistics cannot yet be generated.

```
No simulation has been run yet. Please run a simulation first.
```

Once valid simulation data is available, the system calculates and displays several overall performance indicators. These include the total number of simulation days executed, the total number of customers served across all simulations, the average waiting time, the overall peak hour, the busiest simulation day, and the average teller utilization percentage. These statistic are derived from cumulative counters and aggregated values updated after each simulation cycle.

```
=== Simulation Statistics ===
Total Simulation Days: 3
Total Customers: 314
Average Waiting Time: 0.436306 minutes
Peak Hour (Overall): 09:00 - 10:00
Busiest Day: Thursday (24 customers)
Average Teller Utilization: 36.4861%
```

The peak hour statistic identifies the hour interval during which the largest customer volume was recorded. Similarly, the busiest day identifies the simulation day associated with the highest observed customer count. Teller utilization is calculated using the ratio between total teller working time and total available working time across all simulations. These performance metrics provide insight into the operational efficiency of the simulated banking system and help evaluate how effectively resources are being utilized.

Finally, if the user selects the exit option from the main menu, the system safely terminates execution by setting the program exit condition to true and breaking out of the main control loop. The program then ends normally without data corruption or unexpected behavior.

User Manual

The system allows users to:

- Configure bank settings
- Run queue simulations
- View reports
- Analyze system statistics
- Export reports to text files

1. System Requirements

Software Requirements

- Windows Operating System
- C++ Compiler (Code::Blocks, Dev-C++, Visual Studio, or VS Code with MinGW)
- Console/Terminal support

Required Libraries

The program uses the following C++ libraries:

```
#include <iostream>
#include <fstream>
#include <windows.h>
#include <cstring>
#include <ctime>
#include <cstdlib>
#include <iomanip>
```

2. Program Structure

The system is divided into four main modules:

Module	Description
Configuration	Defines system settings
Simulation	Runs customer queue simulations
Reports	Displays and exports reports
Statistics	Shows overall performance statistics

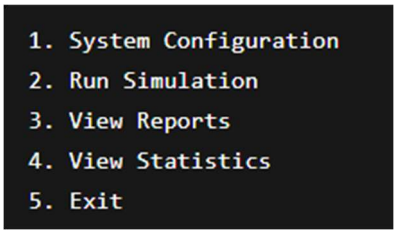
3. Starting the Program

When the program starts, the following screen appears:

```
==========================================
||    BANK CUSTOMERS QUEUING SYSTEM    ||
==========================================
```

The Main Menu will then be displayed.

4. Main Menu



Main Menu Functions

4.1 System Configuration

Used to configure all simulation settings before running the program.

4.2 Run Simulation

Runs the bank queue simulation.

4.3 View Reports

Displays detailed customer and teller reports.

4.4 View Statistics

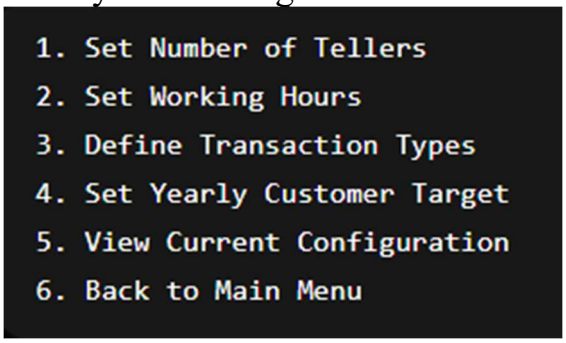
Displays overall statistics collected from simulations.

4.5 Exit

Terminates the program safely.

5. System Configuration

The System Configuration menu contains:



5.1 Set Number of Tellers

This option allows the user to define the number of tellers working each day.

Rules

- Minimum tellers: 1
- Maximum tellers: 10

Example

```
Enter number of tellers for Sunday: 5
```

The system validates:

- Numeric input only
- Value within allowed range

5.2 Set Working Hours

Defines bank opening and closing times.

Format

```
HH:MM
```

Example

```
Enter opening hours: 08:00
Enter closing hours: 16:00
```

Validation

The system checks:

- Correct time format
- Valid hour and minute values
- Closing time must be later than opening time

5.3 Define Transaction Types

Allows the user to define bank services and their average durations.

Example

```
Enter transaction name: Deposit
Enter average duration: 5
```

Rules

- Transaction names must be unique
- Duration must be positive
- Typing done finishes input

Example

```
Deposit
Withdraw
Transfer
done
```

5.4 Set Yearly Customer Target

Defines the estimated yearly number of customers.

Example

```
Enter target number of customers per year: 50000
```

Validation

- Must be numeric
- Must exceed 30,000 customers

The system automatically calculates:
Average Daily Customers = Yearly Target / 365

5.5 View Current Configuration

Displays all current system settings including:

- Number of tellers
- Working hours
- Transaction types
- Yearly target

6. Running Simulations

The simulation menu contains:

```
1. Run Daily Simulation
2. Run Weekly Simulation
3. Run Full Year Simulation
4. Back to Main Menu
```

6.1 Daily Simulation

Simulates one selected day.

Process

The system:

1. Generates random customers
2. Assigns random arrival times
3. Assigns transaction types
4. Sorts customers by arrival time
5. Processes customers using FCFS
6. Calculates waiting times
7. Updates teller activity

Output Example

```
Customers Served: 120
Customers Not Served: 10
Average Waiting Time: 4.2
Peak Hour: 10:00 - 11:00
```

6.2 Weekly Simulation

Runs simulations for all seven days of the week.

Outputs

- Total customers served
- Total customers not served
- Weekly average waiting time
- Most frequent peak hour

6.3 Yearly Simulation

Runs simulations for 365 days.

Outputs

- Total yearly customers
- Customers served/not served
- Average yearly waiting time
- Most common peak hour

7.Queue Logic

The system uses:

First-Come First-Served (FCFS)

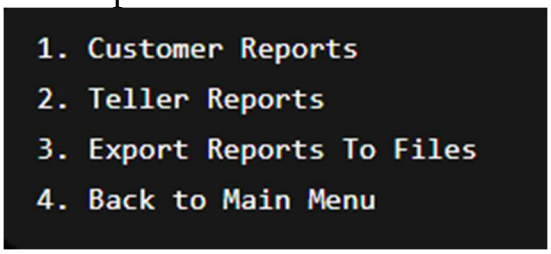
Customers are processed according to arrival time.

Service Logic

- If a teller is available → customer is served immediately
- Otherwise → customer waits
- Long waits may cause customers to leave

8. Reports

The Reports menu includes:



8.1 Customer Reports

Displays:

- Customer ID
- Arrival time
- Service start time
- Service end time
- Transaction type
- Service status

8.2 Teller Reports

Displays:

- Teller ID
- Customers served
- Idle time
- Working time
- Utilization percentage

8.3Export Reports

Exports reports into text files:

File Name	Contents
customers_report.txt	Customer report
tellers_report.txt	Teller report

9.Statistics

The Statistics module displays:

- Total simulation days
- Total customers served
- Average waiting time
- Peak hours
- Busiest day
- Teller utilization

10. Validation and Error Handling

The system includes strong validation to prevent invalid input.

Examples:

Invalid Menu Choice

```
Invalid Choice. Please enter a valid number.
```

Invalid Time Format

```
Invalid opening time. Please enter time in HH:MM format.
```

Invalid Teller Number

```
Please Enter a value between 1 and 10.
```

Invalid Transaction Duration

```
Duration must be positive.
```

.

11. Generated Files

After exporting reports, the following files are created:

```
customers_report.txt
tellers_report.txt
```

These files can be opened using:

- Notepad
- VS Code
- Any text editor

12. Example Workflow

Step 1 — Configure System

- Set tellers
- Set working hours
- Add transaction types
- Set yearly target

Step 2 — Run Simulation

Choose:

- Daily
- Weekly
- Yearly

Step 3 — View Reports

Check customer and teller performance.

Step 4 — View Statistics

Analyze overall bank efficiency.

Code: Link